Physical but not cognitive demand of an exercise bout influences subsequent affective inhibition

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Abstract

Background

The school curriculum places high demands on students' cognitive performance, including the executive domain of inhibitory control. Physical activity breaks have been suggested to restore students' cognitive resources. However, it is unclear which properties (i.e. physical effort and cognitive load) of the physical activity break are suited best to improve subsequent cognitive performance. Therefore, the aim of our study was to investigate the effect of exercise with different physical effort and cognitive load on subsequent inhibitory control.

Methods

Thirty-five healthy right handed children aged 9 to 13 years were recruited from an academic high school in Basel, Switzerland. On four different days at least one week apart and in randomized order, participants underwent short exercise sessions on an interactive exergaming wall (SMARTfit Single, SMARTfit, USA). The sessions lasted 10 min and combined high (or low) physical effort with high (or low) cognitive load. Participants rated their perceived exhaustion and cognitive load using an RPE scale and the NASA task load index. Before and after the exercise sessions, a standard Stroop task and an affective Stroop task were performed to account for changes in inhibitory control with and without an additional component of emotion processing.

Results

The manipulation check showed that the high physical effort condition was related to higher rated perceived exhaustion and the high cognitive load condition to higher scores on the NASA task load index. Statistical analysis using 2 x 2 ANOVA revealed that independent from cognitive load, physical effort influenced reaction time on the affective, but not the standard Stroop task. This indicated a greater improvement following exercise at low compared to high physical effort in both the congruent and incongruent trials. This effect was not due to a speed-accuracy trade-off, given that no main effect of

physical effort was found for accuracy. There were no main effects of cognitive load and no interaction of physical effort and cognitive load.

Conclusion

Our results indicated that in school children, lower exercise intensity is more suitable for improving subsequent inhibitory control than exercise with higher physical effort. However, this effect only seems to be present in situations where cognitive challenge is combined with emotionally charged stimuli. This has implications for recommendations on how to optimally organize students' physical activity during school breaks.

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